



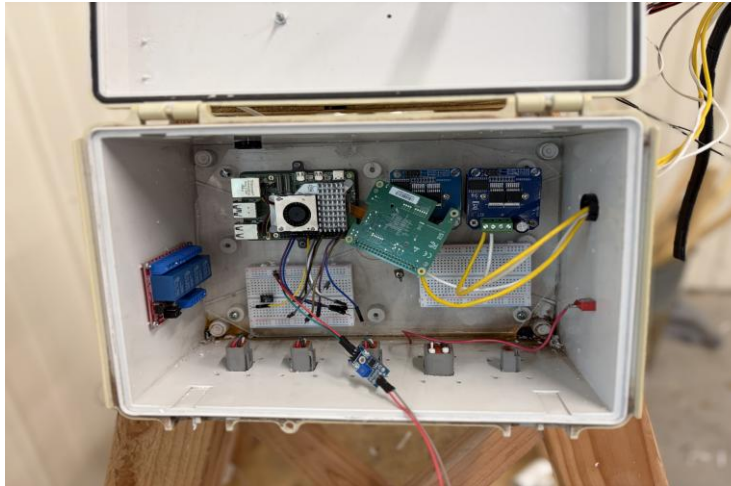
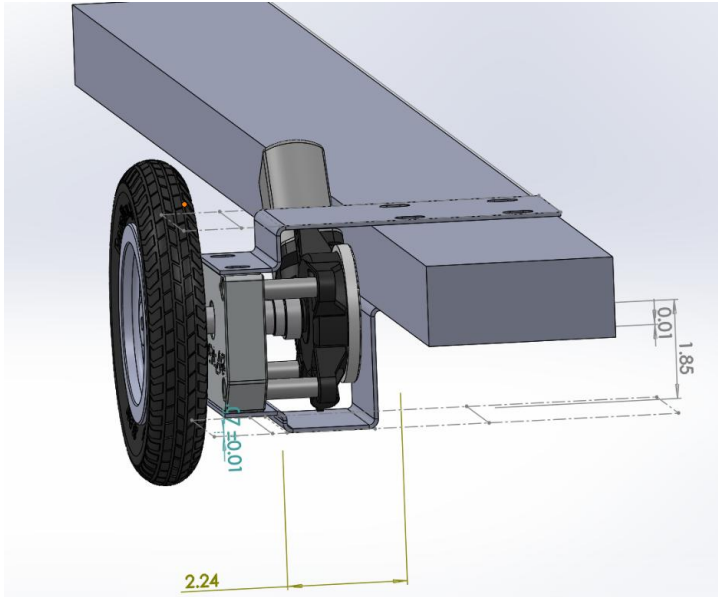
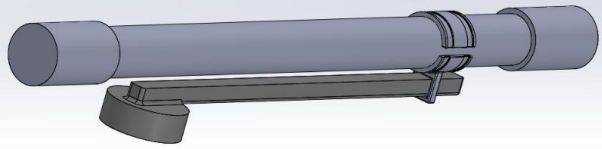
# Smart Center Pivot Irrigation System

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# Overview of project

- Design a working irrigation demonstration platform
- Integrate water control and motion control into one system
- Use sensor inputs to support automated operation
- Provide local operator control through a keypad and OLED interface
- Create a platform that can support data logging, cloud upload, and a cellphone dashboard



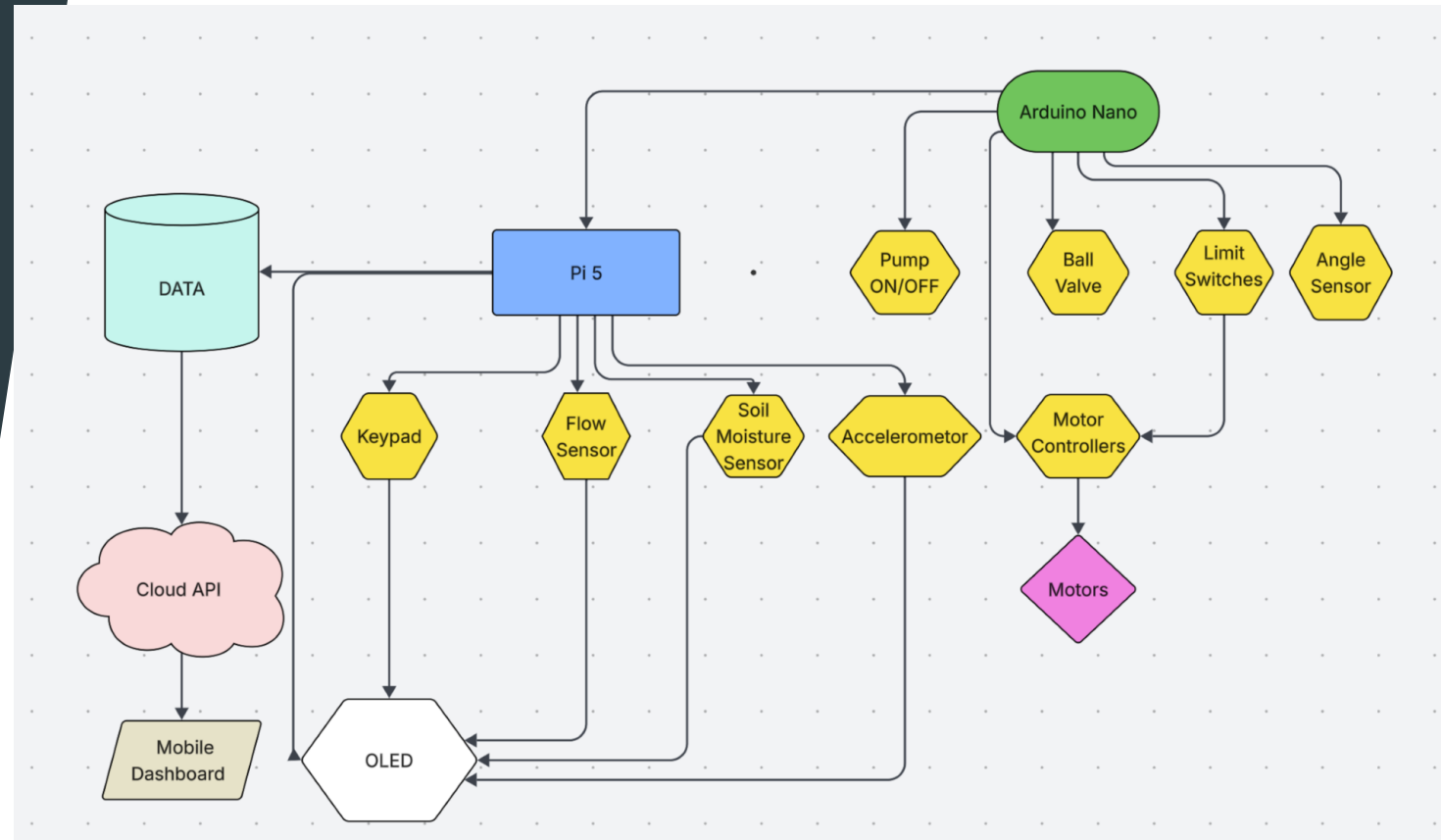


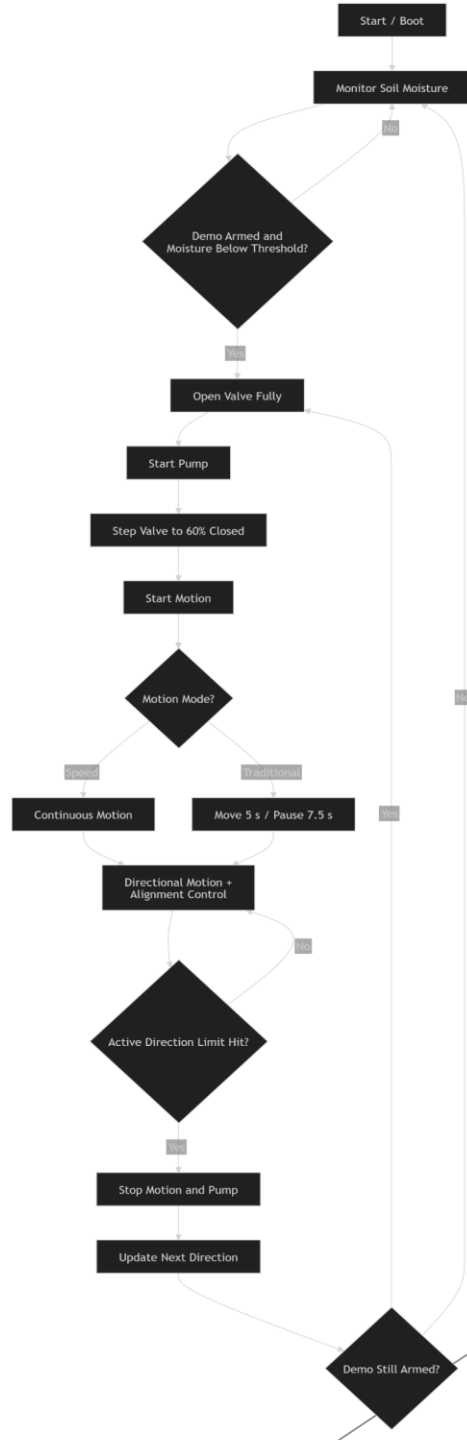
- $\frac{3}{4}$  in pipe
- 16 ft length
- 2 mobile towers
- ~4.5 ft tall
- 5 nozzles
- 12v motors
- Custom mounting brackets
- 3D printed mounts for supports and sensors
- 1 HP pump

# Physical Structure Design

# System Design

- ▶ The system is built around a **Raspberry Pi** and an **Arduino Nano**.
- ▶ The **Raspberry Pi** handles the user interface, local data handling, keypad input, OLED display, and higher-level supervision.
- ▶ The **Arduino Nano** handles real-time local control for motion and water operation.
- ▶ Sensors provide information on soil moisture, flow, vibration, alignment, and travel limits.
- ▶ Pump relays and ball valve relays control water delivery.
- ▶ Motor control logic allows the system to move in either direction while maintaining alignment.





# Control and Logic Systems

## ► Alignment system

- A system that keeps the towers in line while in motion through a magnetic angle sensor and adjusts motor speed based on measured misalignment.

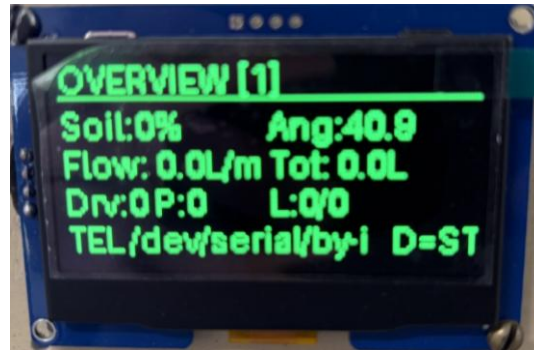
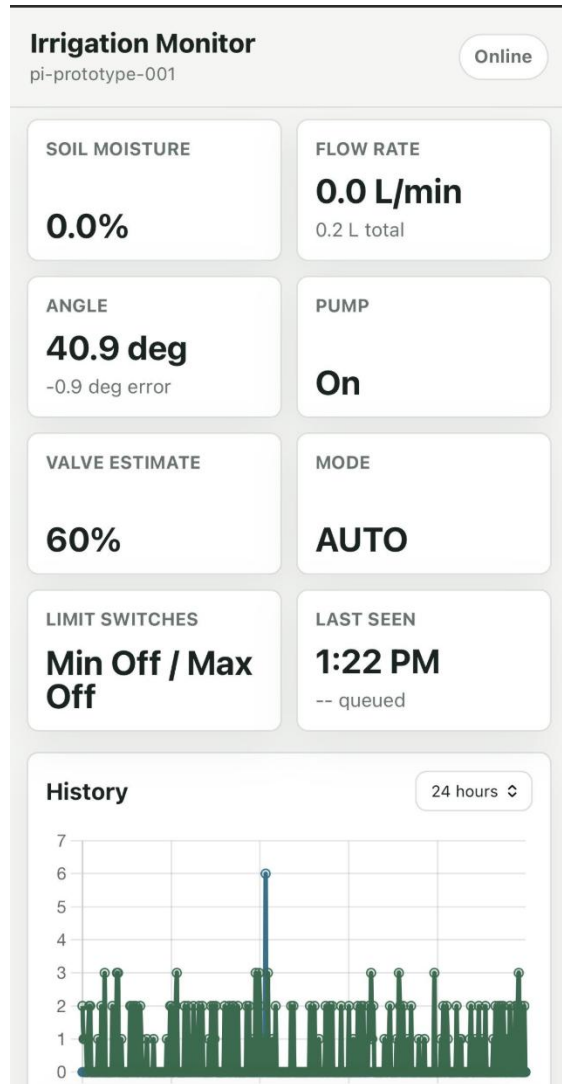
## ► Directional System

- A system run through limit switches that goes 180 degrees and then changes direction after reaching a certain point

## ► Data Logging System

- Our system uses the Raspberry Pi as an edge computer. It collects all sensor data, timestamps it, stores it locally first, and then uploads it to the cloud. The cloud gives us long-term history for dashboards, alerts.

# Mobile Dashboard and OLED Display



## OLED / Keypad Interface

- ▶ Live display of soil moisture, flow, vibration, and controller status
- ▶ Manual pump and ball valve control
- ▶ Drive control with speed, direction, and motion mode selection
- ▶ Demo mode arming, threshold adjustment, and status display

# Results



The current prototype successfully demonstrates:

- local control of pump, ball valve, and motion
- sensor-based irrigation sequence triggering
- directional limit switch behavior
- automatic demo rearming
- selectable motion modes
- local user interface through keypad and OLED
- cloud logging and cellphone dashboard

